

Tutorial 3

Coverage: Unit 3

Question 1

Lando and Oscar are both freelance social media managers. A new platform launches, effectively doubling their hourly rate. Lando decides that since he is earning so much more, he can finally afford to take Friday afternoons off to go hiking. Oscar decides that since every hour is now worth so much more, he will cancel her weekend plans to take on extra clients.

Use the concepts of Income Effect and Substitution Effect to explain their behaviours. In your answer, identify which effect dominates for Lando and which dominates for Oscar.

Question 2: True and False Questions

- a) Any outcome (i.e. any combination of two consumption goods) that emerges from decision-making under scarcity will be an optimal choice.

- b) An indifference curve (IC) shifts outwards when the budget increases.

- c) At the optimal consumption point, the Marginal Rate of Substitution (MRS) represents the "price" the consumer is willing to pay, while the Marginal Rate of Transformation (MRT) represents the "price" the market demands.

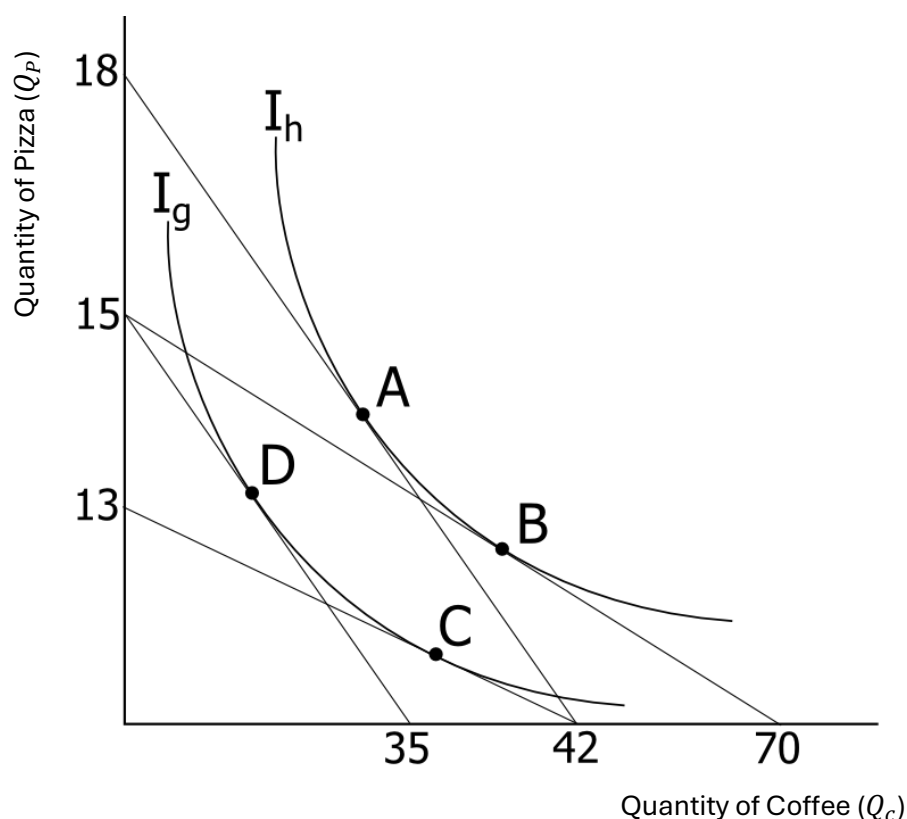
Question 3

Kanya has a total of 10 hours a day to split between working and watching Netflix (free time). She can earn R150 per hour doing private tutoring. Let t be her hours of Netflix and C be her total daily consumption in Rand.

- 3.1 What is the equation for Kanya's budget constraint?
- 3.2 If Kanya decides she needs exactly R600 a day for her expenses, how many hours of Netflix can she afford?

Question 4

Lily is a consumer who earns an income of R630 per week. She uses her entire budget to purchase two goods: pizza and coffee, which are both normal goods. Several indifference curves (I_g and I_h) and budget constraints (the straight lines) are indicated in the figure below. Lily currently consumes at her optimal point, **D**.



- 4.1 What are the current prices of a coffee and a pizza?
- 4.2 Write down an equation that represents her current budget constraint as represented in the figure above [**HINT**: You can start off with the equation:

$$\text{Total income} = \text{Total consumption}$$

$$630 = (P_c \times Q_c) + (P_p \times Q_p)$$

and rearrange to have Q_p on its own on either side of the equation].

- 4.3 The price of coffee drops by 50%. Her new budget constraint is indicated on the figure above. Identify (using the points labelled **A**, **B**, **C** and **D**) the following effects in the figure:
- 4.3.1 Income effect
- 4.3.2 Substitution effect
- 4.3.3 Overall (price) effect